

January 12, 2006



U.S. Department
of Transportation

400 Seventh Street, S.W.
Washington, D.C. 20590

**Pipeline and
Hazardous Materials
Safety Administration**

DOT-SP 9831
(FIFTH REVISION)

EXPIRATION DATE: December 31, 2009

(FOR RENEWAL, SEE 49 CFR § 107.109)

1. GRANTEE: L'Air Liquide, Sassenage, France
(U.S. Agent: Air Liquide America L.P.
Houston, TX)
2. PURPOSE AND LIMITATIONS:
 - a. This special permit authorizes the manufacture, mark, sale and use of a vacuum insulated non-DOT specification portable tank to be used for the transportation in commerce of liquefied helium. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR §§ 172.203(a), 173.318, 176.30, 176.76(g)(1) in that a non-DOT specification insulated portable tank is not authorized, except as specified herein.
5. BASIS: This special permit is based on the application of Air Liquide America L.P. on behalf of L'Air Liquide dated, August 30, 2005, submitted in accordance with § 107.109.

6. HAZARDOUS MATERIALS (49 CFR § 172.101):

Proper Shipping Name/ Hazardous materials description	Hazard Class/ Division	Identi- fication Number	Packing Group
Helium, refrigerated liquid (<i>cryogenic liquid</i>)	2.2	UN1963	N/A

7. SAFETY CONTROL MEASURES:

a. Prescribed packaging is a vacuum insulated non-DOT specification portable tank designed and constructed in accordance with Section VIII of the ASME Code and subparagraph b. of this paragraph. The portable tank is enclosed in an ISO type frame. Design pressure is 42 PSIG for the internal tank. The portable tank is vacuum insulated with a helium shield. Design temperature is -452°F. Water capacity is 1,050 gallons, nominal for the inner tank. Tank material is SA 240 Type 304L stainless steel for the inner tank, and austenitic stainless steel for the helium insulation tank and the outer jacket.

b. Each portable tank must conform to L'Air Liquide drawings nos. 649 3451-A0 649 3451-A0 002 Rev. C dated June 16, 1986, 649 3451-A0 004 Rev. A dated May 28, 1985, 649 3541-A0 005 Rev. A dated May 28, 1985, 649 3451-A0 007 Rev. 0, 649 3451-A0 008 Rev. B dated June 30, 1986, 649 3451-A0 010 Rev. B dated June 25, 1986, other reference drawings, calculations and technical specifications on file with the Office of Hazardous Materials Exemptions and Approvals (OHMEA) and with § 178.338 except as follows:

(1) Impact testing is required for stainless steel tanks with a design temperature colder than -425°F.

(2) Section 178.338-10 does not apply.

(3) The portable tank need not conform with § 178.338-13(b) or (c). Lifting lugs, framework and any anchoring to the inner tank, the helium shield tank or the tank jacket must conform with § 178.338-13(a). Each portable tank design must be qualified in accordance with § 178.270-13(c).

(4) "DOT-SP 9831" must replace the mark "MC 338".

8. SPECIAL PROVISIONS:

a. Each portable tank must be reinspected and retested once every 5 years at a pressure (in PSIG) of one and one-fourth times the sum of the design pressure plus the static head plus 14.7 psig in accordance with § 180.605 as prescribed for DOT 51 Specification portable tanks. The test pressure must be reduced by 14.7 psig if the inner tank is under vacuum while under test.

b. Each portable tank must be plainly marked "DOT-SP 9831" on both sides near the middle, in letters at least two (2) inches high on a contrasting background.

c. Each portable tank must be prepared and shipped as required in § 173.318, as applicable for the lading.

d. Shipments by cargo vessel must conform with the following:

(1) The package must conform with the requirements of § 176.76(h). The portable tank may not be overstowed with other containers or freight.

(2) The legend "One-Way Travel Time _____ Hours" or "OWTT _____ Hours" must be marked on the shipping paper and on the dangerous cargo manifest immediately after the container description. The OWTT is determined by the formula:

OWTT = MRHT - 24 hours.

(3) A written record of the portable tank's pressure and ambient (outside) temperature at the following times must be prepared for each shipment.

(i) At the start of each trip;

(ii) Immediately before and after any manual venting;

(iii) At least every 24 hours; and

(iv) At the destination point.

January 12, 2006

(4) Any lading road relief valve set at a pressure lower than that prescribed for the (safety) pressure relief valve must be closed during transportation by cargo vessel.

e. No person may transport or offer for transportation a charged portable tank unless the pressure of the lading is equal to or less than that used to determine the marked rated holding time and the OWTT is equal to or greater than the elapsed time between the start and termination of travel.

f. The actual holding time for each tank be determined after each shipment. If it is determined that the actual holding time is less than 90 percent of the MRHT of the tank, the tank may not be refilled until it is restored to its MRHT or the tank is remarked with the holding time determined by this examination.

g. The portable tank must be secured to the motor vehicle in accordance with the requirements of 49 CFR §§ 393.100 through 393.106. Additionally, the motor vehicle's bumper must be located at least 6 inches to the rear of any tank component used for loading or unloading that may contain lading during transit.

h. In accordance with the provisions of Paragraph (b) of § 173.22a, persons may use the packaging authorized by this special permit for the transportation of the hazardous materials specified in paragraph 6, only in conformance with the terms of this special permit.

i. A person who is not a holder of this special permit, but receives a package covered by this special permit, may reoffer it for transportation provided no modification or change is made to the package or its contents and it is offered for transportation in conformance with this special permit and the HMR.

j. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.

k. Each packaging manufactured under the authority of this special permit must be either (1) marked with the name of the manufacturer and location (city and state) of the facility at which it is manufactured or (2) marked with a registration symbol designated by the Office of Hazardous

January 12, 2006

Materials Exemptions and Approvals for a specific manufacturing facility.

1. A current copy of this special permit must be maintained at each facility where the package is manufactured under this special permit. It must be made available to a DOT representative upon request.
- m. Packagings permanently marked 'DOT-E 9831', prior to October 1, 2007 may continue to be used under this special permit for the remaining service life of the packaging or until the special permit is no longer valid. Packagings marked on or after October 1, 2007 must be marked 'DOT-SP 9831'.
9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle and cargo vessel.
10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each cargo vessel or motor vehicle used to transport packages covered by this special permit.
11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:
- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
 - o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
 - o Registration required by § 107.601 et seq., when applicable.

Each "Hazmat employee", as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU)—"The Hazardous Materials Safety and Security Reauthorization Act of 2005" (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term "exemption" to "special permit" and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.



for Robert A. McGuire
Associate Administrator for
Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Materials Safety Administration, Department of Transportation, Washington, D.C. 20590. Attention: PHH-31.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at http://hazmat.dot.gov/sp_app/special_permits/spec_perm_index.htm Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: kah